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1	2	3	4	5	6	7	8
A	UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-25 & MVD-26 20kV Centralized Panel Room MNA SCHEMATIC DIAGRAM SBC-W INCOMING H01,H10						A
B							B
C							C
D							D
E							E
F							F
00	Issued for Approval	15.01.19	YR	Checked	AW		
Ind.	Revision	Date	Name	Approved	AI		
1	2	3	4	5	6	7	8

PROJECT TITLE		COVER SHEET		Scale	NTS	= H01,H10
MV Switchgear for MNA Serang Plant		MVD-25 & MVD-26 20kV Centralized Panel Room MNA		Customer	PT Multimas Nabati Asahan	& 304
PT. ABB Sakti Industri		SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-304.07	Sh. 1
Based on: BE318042		SBC-W INCOMING		Customer Doc. Nbr.		Cont. 28

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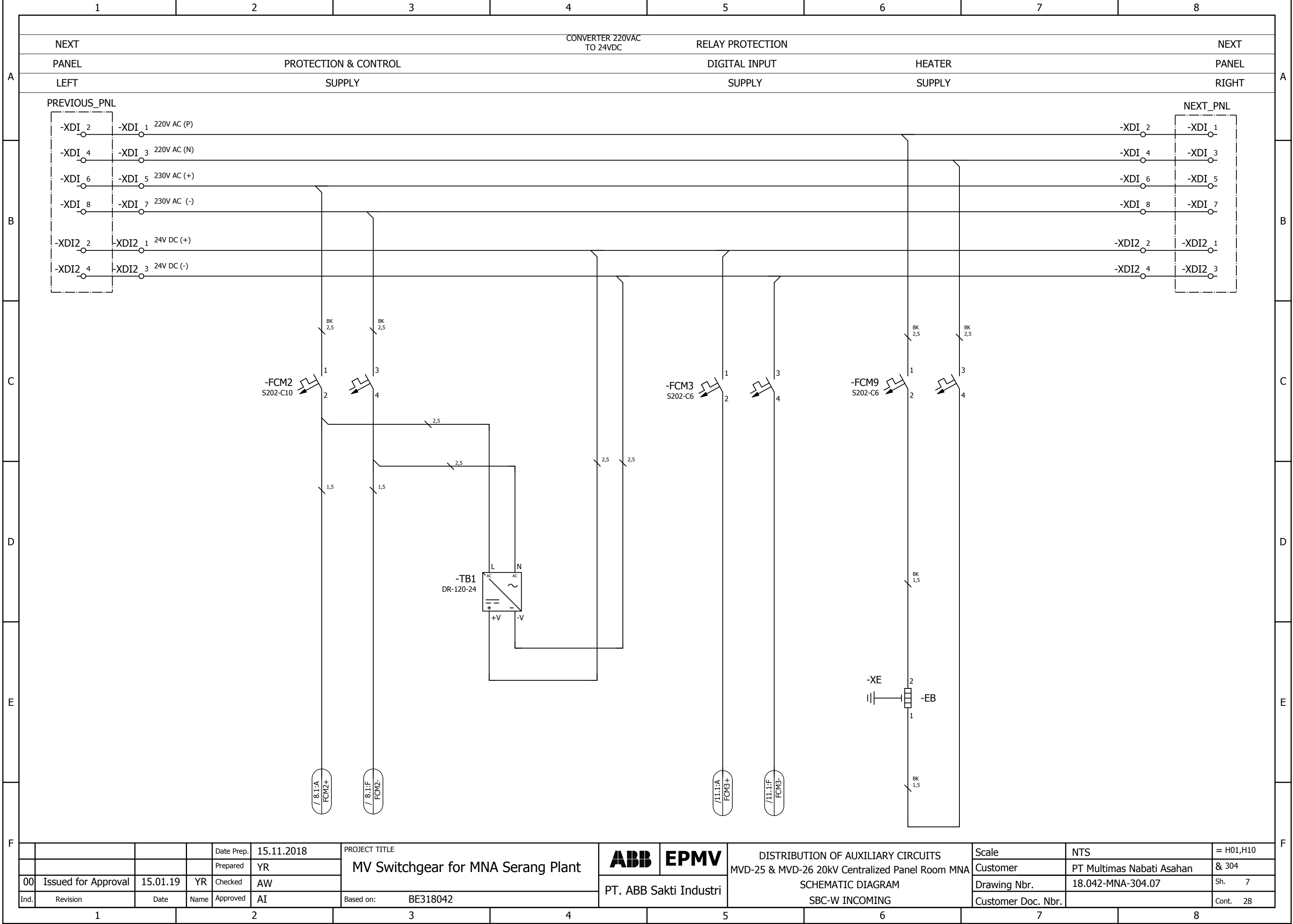
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A	SHEET INDEX								A						
	PAGE			PAGE NAME				REVISION							
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	==MVD-25_MVD-26=H01,H10&304/2			Index of Sheet				00							
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	==MVD-25_MVD-26=H01,H10&304/4			REFERENCE DESIGNATIONS				00							
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	==MVD-25_MVD-26=H01,H10&304/6			REFERENCE DESIGNATIONS				00							
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	==MVD-25_MVD-26=H01,H10&304/16			LV DOOR LAYOUT				00							
	==MVD-25_MVD-26=H01,H10&304/17			LV BOX LAYOUT				00							
	==MVD-25_MVD-26=H01,H10&304/18			PART LIST				00							
	==MVD-25_MVD-26=H01,H10&304/19			Terminal diagram ==MVD-25_MVD-26=H01,H10+-XDA											
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	==MVD-25_MVD-26=H01,H10&304/21			Terminal diagram ==MVD-25_MVD-26=H01,H10+-XDC1											
	==MVD-25_MVD-26=H01,H10&304/22			Terminal diagram ==MVD-25_MVD-26=H01,H10+-XDC2											
	==MVD-25_MVD-26=H01,H10&304/23			Terminal diagram ==MVD-25_MVD-26=H01,H10+-XDI											
	==MVD-25_MVD-26=H01,H10&304/24			Terminal diagram ==MVD-25_MVD-26=H01,H10+-XDI2											
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	==MVD-25_MVD-26=H01,H10&304/27			Terminal diagram ==MVD-25_MVD-26=H01,H10+-XDV											
	==MVD-25_MVD-26=H01,H10&304/28			Terminal diagram ==MVD-25_MVD-26=H01,H10+-XDX											
E									E						
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet		Scale	NTS	= H01,H10
				Prepared	YR	MV Switchgear for MNA Serang Plant		MVD-25 & MVD-26 20kV Centralized Panel Room MNA			Customer	PT Multimas Nabati Asahan	& 304		
	00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-304.07	Sh. 2
	Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				SBC-W INCOMING		Customer Doc. Nbr.		Cont. 28
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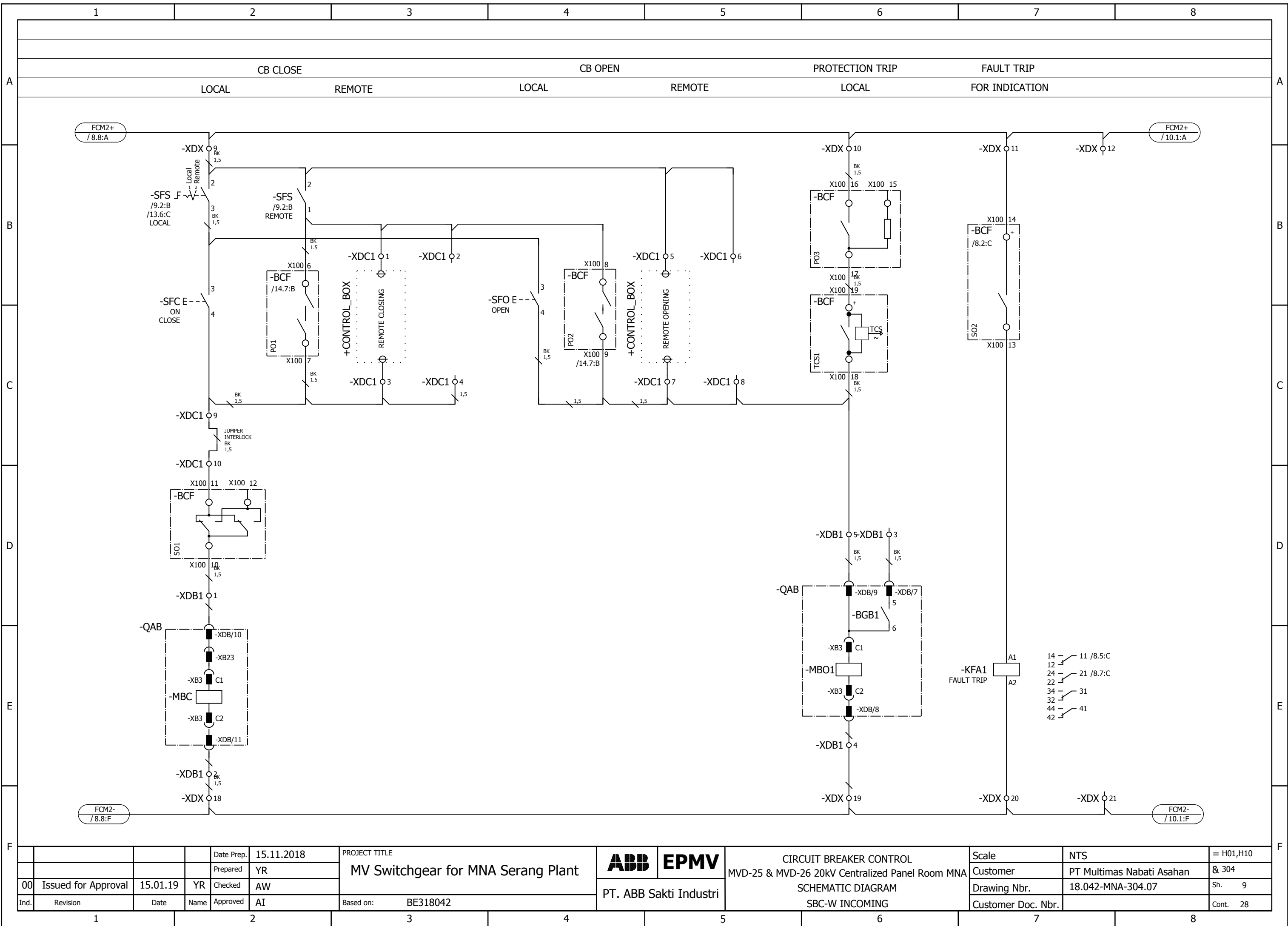
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A	-TFA	ACTIVE POWER TRANSDUCERS						-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS							A
	-TFC	CURRENT TRANSDUCERS						-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING							
	-TFF	FREQUENCY TRANSDUCERS						-XDI3	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION							
	-TFJ	ACTIVE ENERGY TRANSDUCERS						-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS							
	-TFK	REACTIVE ENERGY TRANSDUCERS						-XDI6	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS							
B	-TFM	MULTIFUNCTION TRANSDUCERS						-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS							B
	-TFP	POWER-FACTOR TRANSDUCERS						-XDI8	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS							
	-TFQ	REACTIVE POWER TRANSDUCERS						-XDI9	TERMINAL BLOCK FOR SPECIAL USE							
	-TFS	SIGNAL CONVERTERS						-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD							
	-TFV	VOLTAGE TRANSDUCERS						-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS							
C	-WA	BUSBARS						-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT							C
	-WBC	POWER CABLES						-XDS	CURRENT SOCKETS							
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.						-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB							
	-WGC	CONTROL CABLES						-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK							
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)						-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK							
D	-WH	OPTICAL FIBRES						-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS							D
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS						-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1							
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1						-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2							
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2						-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3							
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3						-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR							
E	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER						-XDX	SUPPORT TERMINAL BLOCK							E
	-XDA8	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS						-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE							
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)						-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE							
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER						-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE							
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS						-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE							
F	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS						-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE							F
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE							
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT							
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR														
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET														
G	-XDC	CUSTOMER TERMINAL BLOCK														G
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER														
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR														
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT														
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS														
H	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER														H
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE														
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE														
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2														
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2														
I	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS														I
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR														
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS														
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS														
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)														
J	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)														J

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant				REFERENCE DESIGNATIONS MVD-25 & MVD-26 20kV Centralized Panel Room MNA		Scale	NTS		= H01,H10
				Prepared	YR							Customer	PT Multimas Nabati Asahan		& 304
00	Issued for Approval	15.01.19	YR	Checked	AW							Drawing Nbr.	18.042-MNA-304.07		Sh. 6
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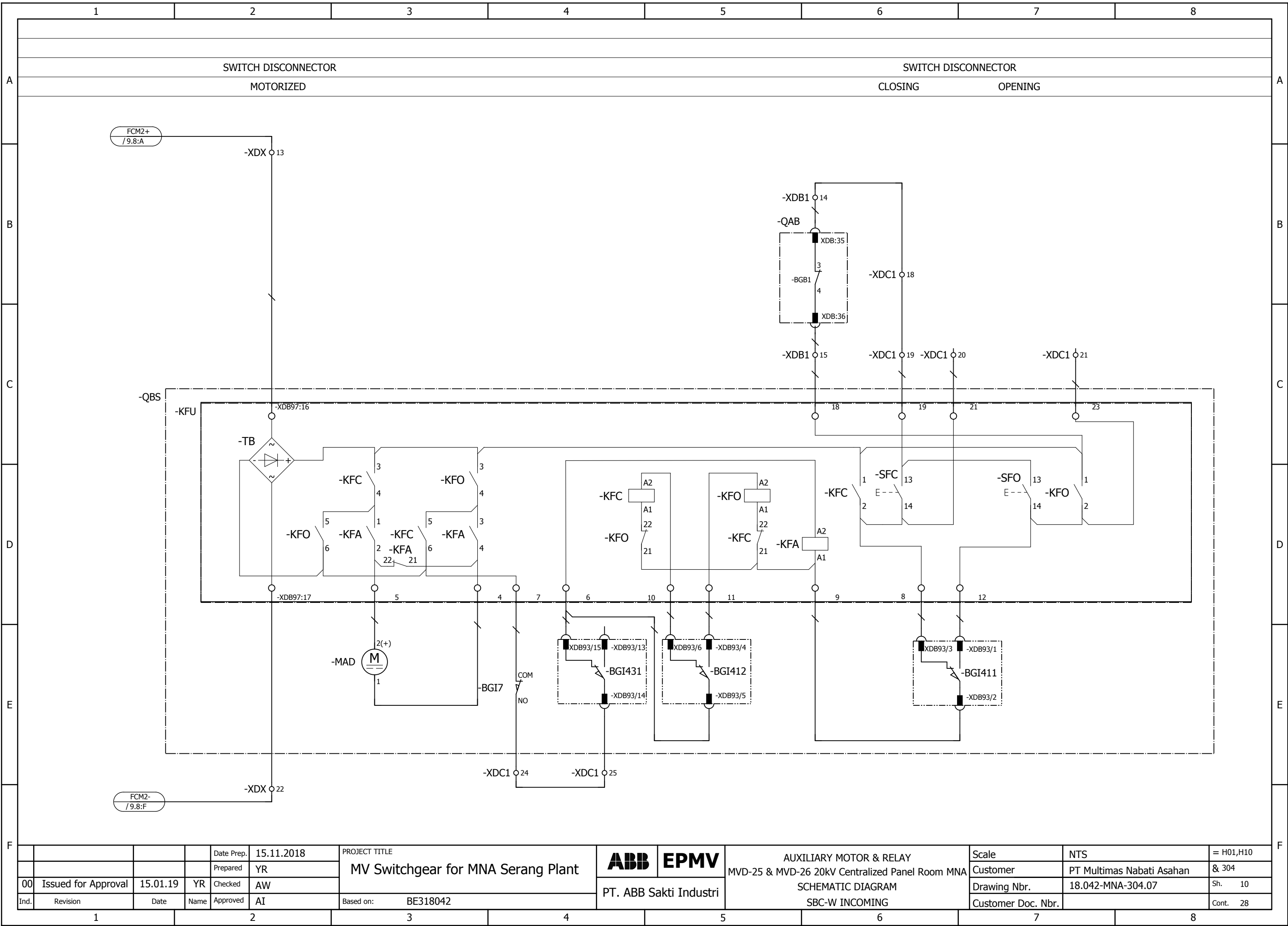
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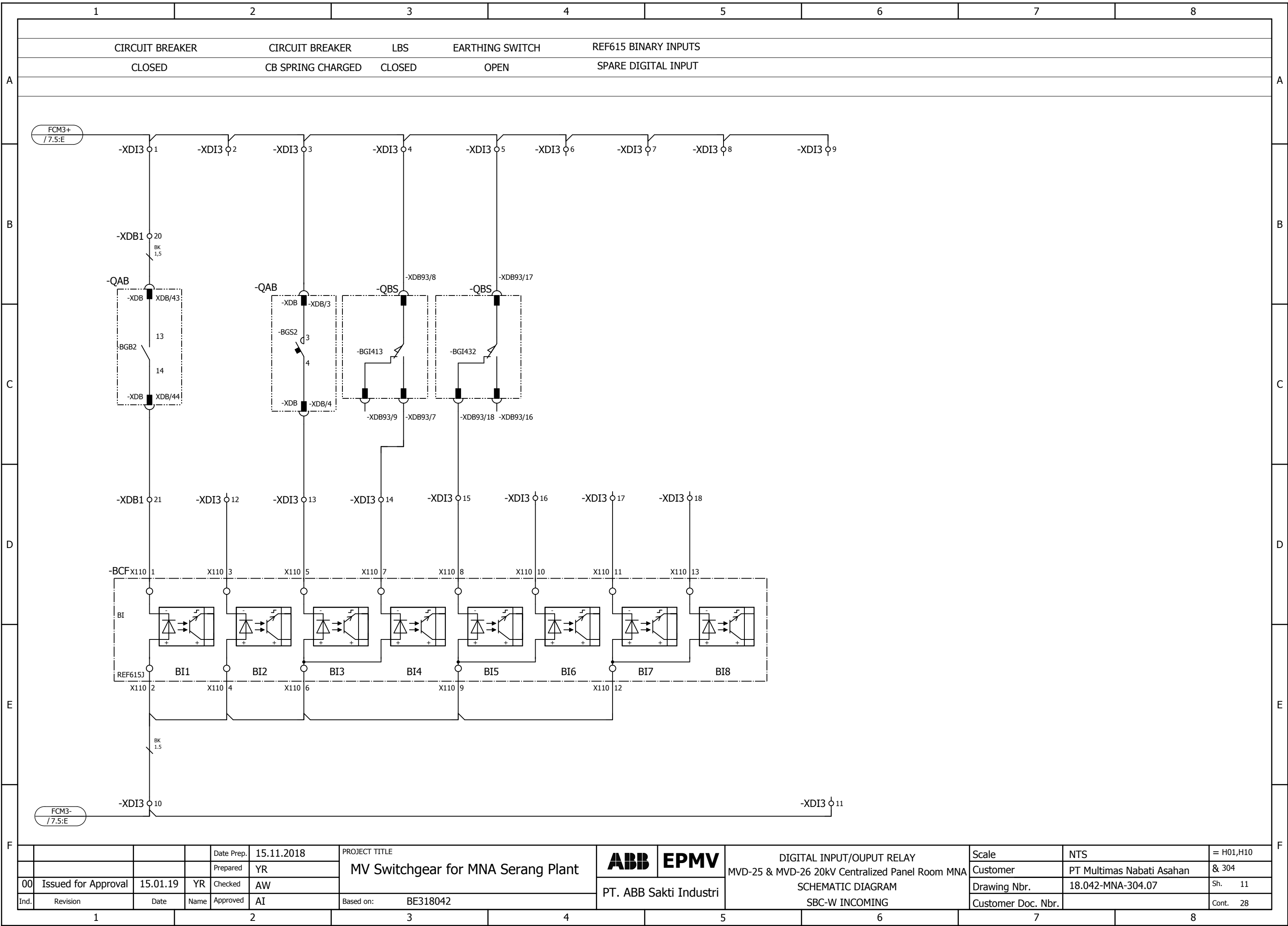
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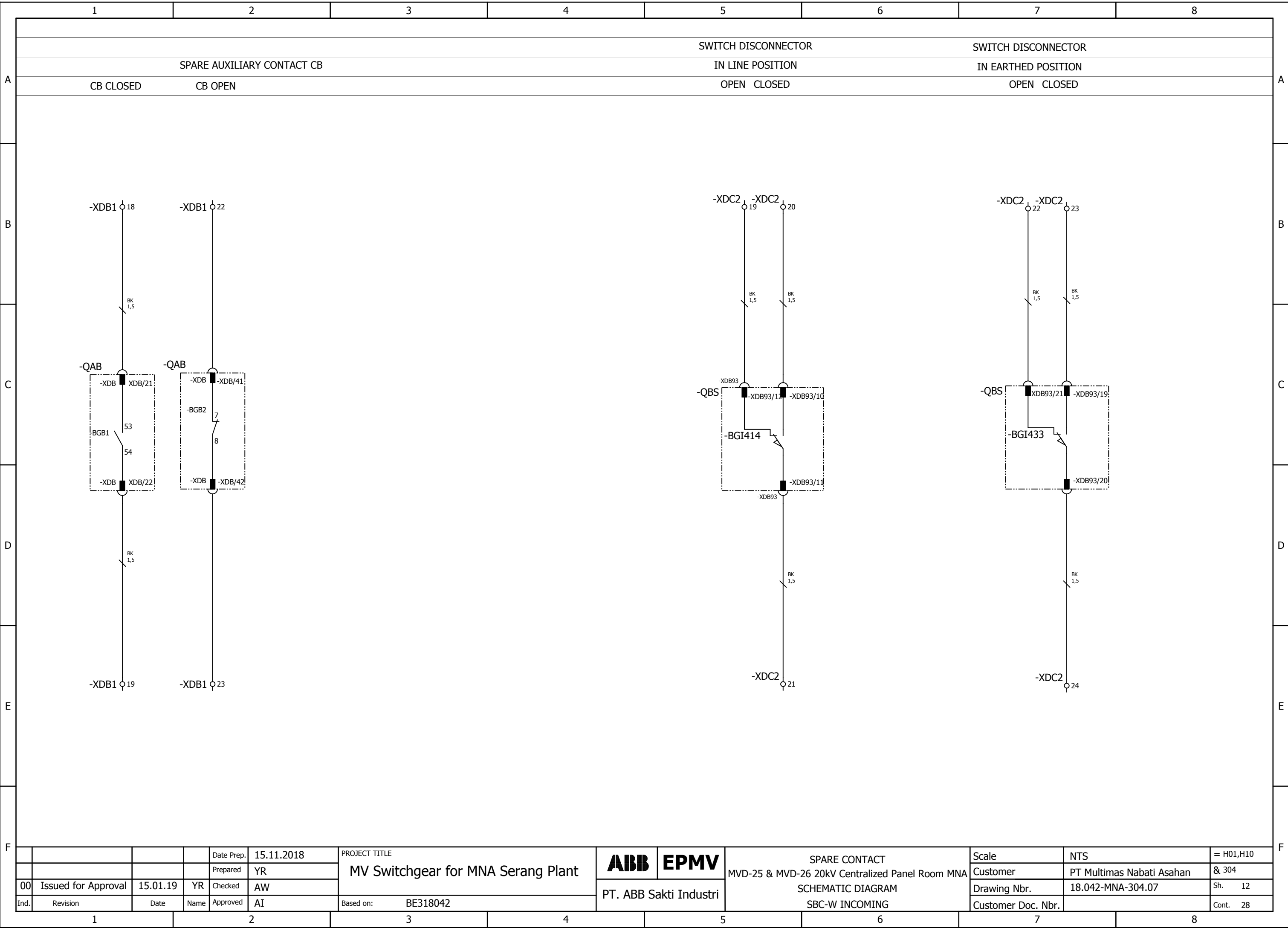


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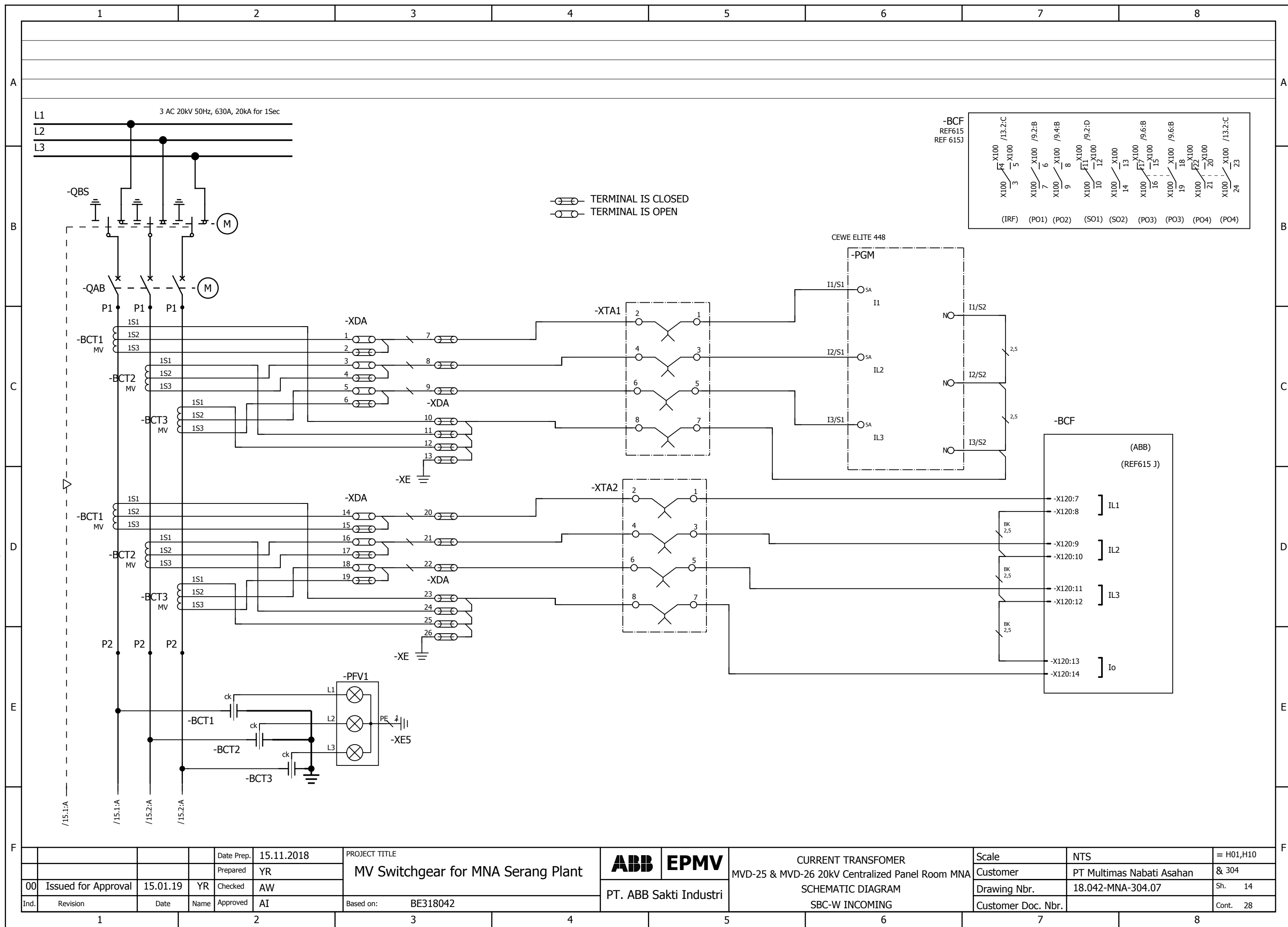
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	DIGITAL INPUT/OUTPUT RELAY MVD-25 & MVD-26 20kV Centralized Panel Room MNA SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01,H10
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.042-MNA-304.07	Sh.	11
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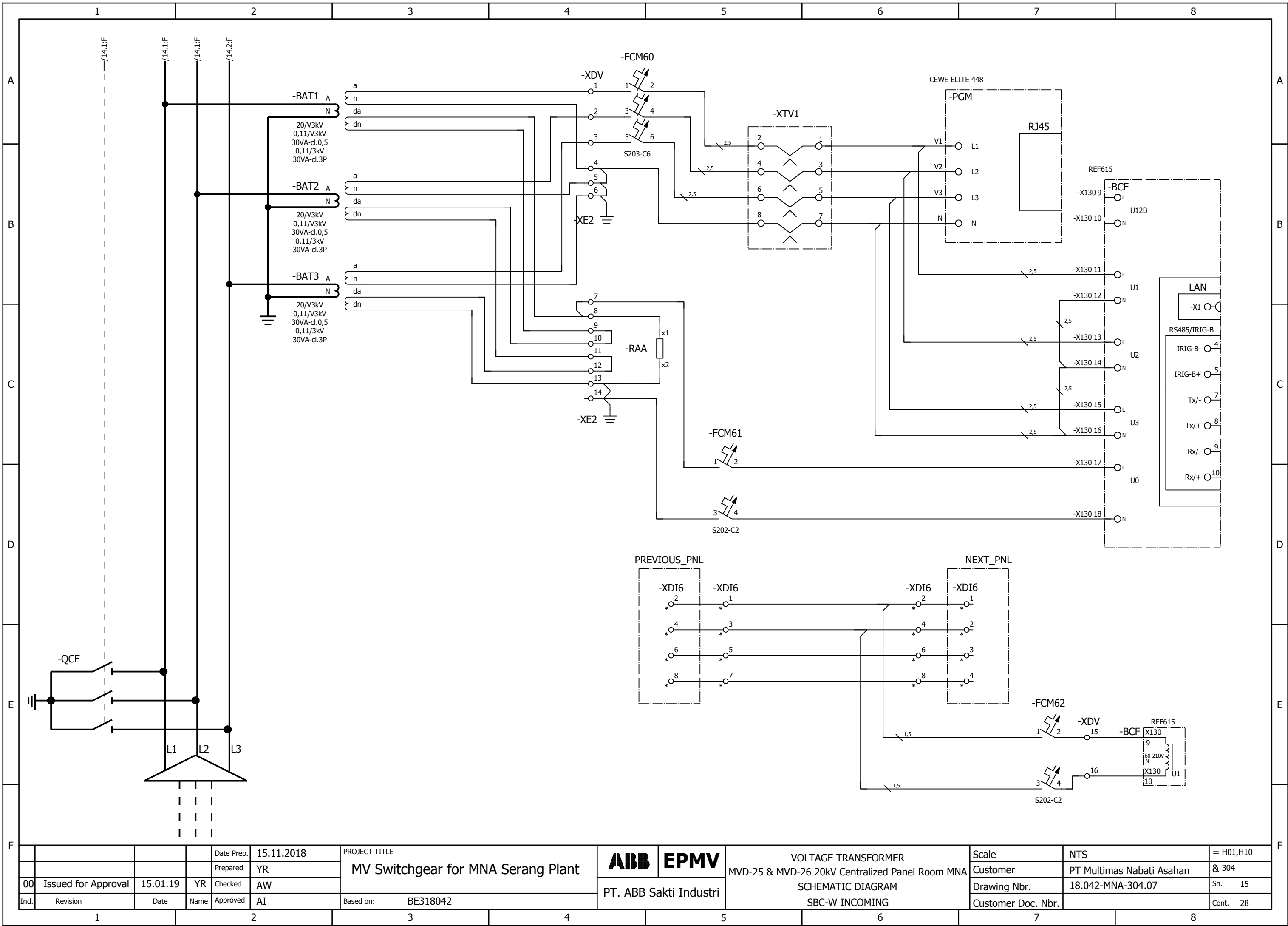


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SPARE CONTACT MVD-25 & MVD-26 20kV Centralized Panel Room MNA SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01,H10
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.042-MNA-304.07	Sh. 13	
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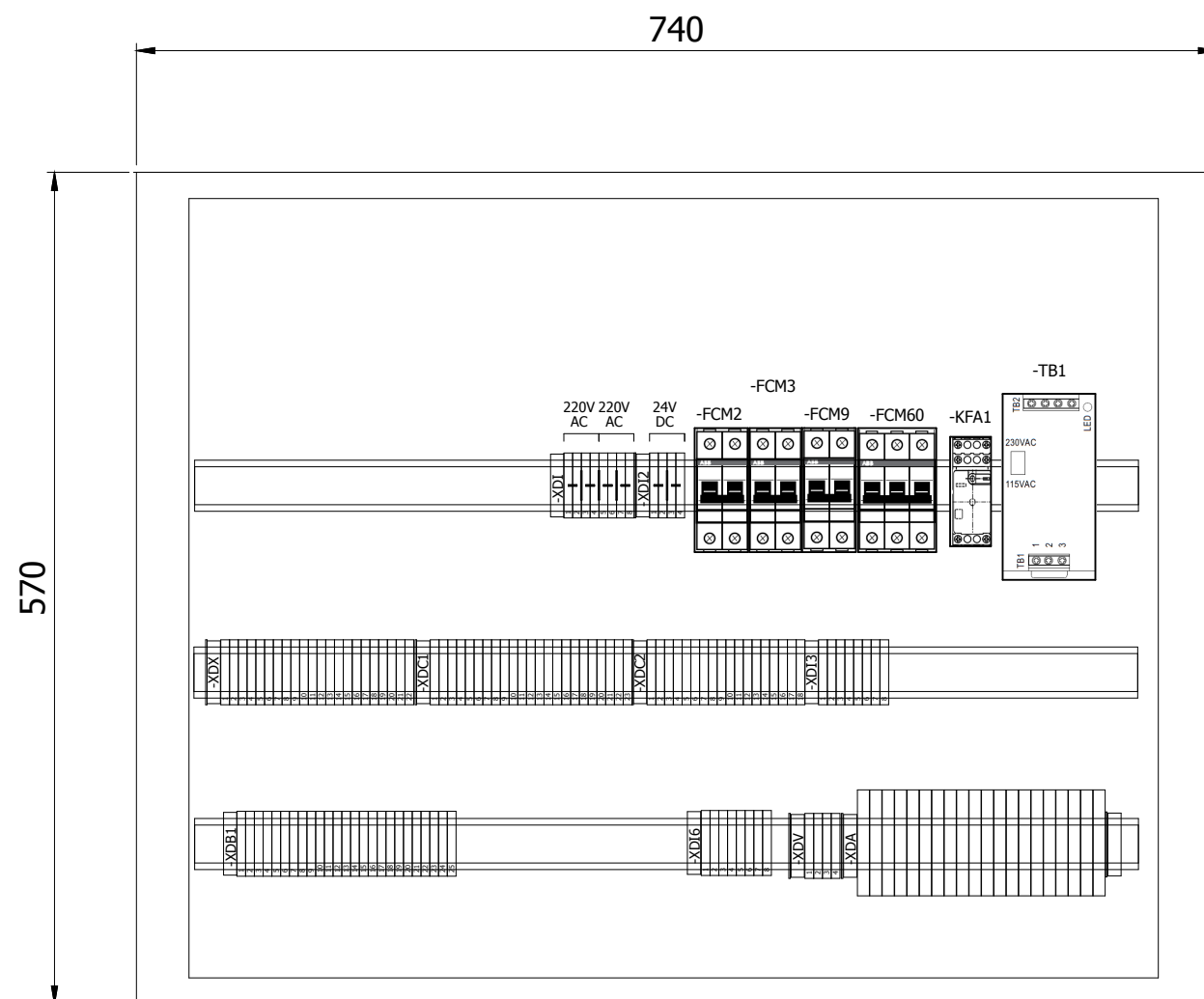
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No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK
10	-XTV1 (TEST BLOCK , YONGSUNG)	-XTV, VOLTAGE TEST BLOCK

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB EPMV	LV BOX LAYOUT MVD-25 & MVD-26 20kV Centralized Panel Room MNA SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01,H10
				Prepared	YR				Customer	PT Multimas Nabati Asahan	& 304
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Summarized parts list														IDABB_F02_005-IDABB													
Carry over: 0.00																											
A	TAG ID		DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS		A												
	-BCF		Feeder Management Relay HBFJAEAGABC1ANA11G				ABB		REF615J HBFJAEAGABC1ANA11G		2		/8.2:C;/15.7:B														
B	-EB		HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1		/7.6:E		B												
	-FCM2		MCB S202-C10 MCB. 2P, 10A				ABB		2CDS252001R0104		1		/7.2:C														
	-FCM3;-FCM9		MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		2		/7.5:C;/7.6:C														
	-FCM60		MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1		/15.4:A														
	-FCM61;-FCM62		MCB S202-C2 MCB. 2P, 2A				ABB		2CDS252001R0024		2		/15.5:D;/15.7:E														
C	-KFA1		Auxiliary Relay 230AC 4COS				PHOENIX		1YRA3DL266230001		1		/9.7:E		C												
	-PFG1		PILOT LIGHT GREEN #CL-523G#230VAC CL-523G				ABB		1SFA619402R5232		1		/8.5:E														
	-PFR1		PILOT LIGHT RED #CL-523R#230VAC CL-523R				ABB		1SFA619402R5231		1		/8.4:E														
	-PFY1		PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y				ABB		1SFA619402R5233		1		/8.6:E														
	-PGM1		POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC				CEWE		1YRA4EA0350001		1		/8.3:C														
D	-SFC		PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR				IDEC		1YRA3DA14700001		1		/9.2:B		D												
	-SFO		PUSH BUTTON, YW1B-M1E10-R RED COLOUR				IDEC		1YRA3DA14700002		1		/9.4:B														
	-SFS		LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)				K&N		1YRA3DA17940011		1		/9.2:B														
	-TB1		Voltage Input : 85-264 VAC/120-370V DC Voltage Output : 24 VDC, 5A				MEANWELL		DR-120-24		1		/7.3:D														
	-XTA1;-XTA2		Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSCTT-04C		2		/14.4:B;/14.4:D														
E	-XTV1		Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSPTT-04C		1		/15.5:A		E												
F					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	PART LIST MVD-25 & MVD-26 20kV Centralized Panel Room MNA		Scale		NTS	= H01,H10	F										
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